

Reason with number properties

Use the scaffold, show each step and ask for an adult check when marked.

Name: _____ Date: _____

1. Read the model: Find the smallest number divisible by 6, 8 and 9. Copy or complete the first step.

2. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

3. Which prime factors make 84?

4. Miro says: Miro assumes every odd number is prime. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: Find the smallest number divisible by 6, 8 and 9. Copy or complete the first step.

Answer 72.

2. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

Answer 1 x 96, 2 x 48, 3 x 32, 4 x 24, 6 x 16 and 8 x 12.

3. Which prime factors make 84?

Answer $2 \times 2 \times 3 \times 7$.

4. Miro says: Miro assumes every odd number is prime. Is this correct? Explain.

Answer Odd composite numbers have factors other than 1 and themselves, for example 9 and 15.

Reason with number properties

Work independently. Show a complete method and check at least one answer.

Name: _____ Date: _____

1. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.

2. Which prime factors make 84?

3. Find two prime numbers with a sum of 60 and a difference of 14.

4. Identify and correct this misconception: Miro assumes every odd number is prime.

5. Write a complete sentence explaining how you checked one answer.

Teacher answers and checking prompts

1. A rectangle has area 96 square units and whole-number sides. Give every possible side pair.
Answer 1 x 96, 2 x 48, 3 x 32, 4 x 24, 6 x 16 and 8 x 12.
2. Which prime factors make 84?
Answer 2 x 2 x 3 x 7.
3. Find two prime numbers with a sum of 60 and a difference of 14.
Answer 23 and 37.
4. Identify and correct this misconception: Miro assumes every odd number is prime.
Answer Odd composite numbers have factors other than 1 and themselves, for example 9 and 15.
5. Write a complete sentence explaining how you checked one answer.
Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

Reason with number properties

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: _____ Date: _____

1. Find two prime numbers with a sum of 60 and a difference of 14.

2. Explain why this reasoning fails: Miro assumes every odd number is prime.

3. Create a new example that tests the same idea as: Which prime factors make 84?

4. Find a second method or representation. Compare its efficiency with your first method.

5. Write one always, sometimes or never statement about today's learning and justify it.

Teacher answers and checking prompts

1. Find two prime numbers with a sum of 60 and a difference of 14.
Answer 23 and 37.
2. Explain why this reasoning fails: Miro assumes every odd number is prime.
Answer Odd composite numbers have factors other than 1 and themselves, for example 9 and 15.
3. Create a new example that tests the same idea as: Which prime factors make 84?
Answer Answers vary. The example and solution must preserve the mathematical structure.
4. Find a second method or representation. Compare its efficiency with your first method.
Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.
5. Write one always, sometimes or never statement about today's learning and justify it.
Answer Answers vary. A valid example and counterexample should support the classification.